ex de pe curs

1.

Exercise 3

Consider training a classification decision tree given a design matrix $\boldsymbol{X} = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $\boldsymbol{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of $\boldsymbol{X}$, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest Gini index: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

Solution:

- the *Gini index* is defined by:

$$G_h = \sum_{k=1}^{q} \hat{p}_{hk}(1 - \hat{p}_{hk}) = \sum_{k=1}^{q} \hat{p}_{hk} - \sum_{k=1}^{q} \hat{p}_{hk}^2 = 1 - \sum_{k=1}^{q} \hat{p}_{hk}^2,$$

where $\hat{p}_{hk}$ represents the proportion of training examples in the hth region that are from the kth class:

$$\hat{p}_{hk} = \frac{1}{|\mathcal{D}_h|} \sum_{i:\,(\boldsymbol{x}^{(i)}, y^{(i)}) \in \mathcal{D}_h} 1_{y^{(i)}=k}.$$

$q = 2 \longrightarrow$ nr clase

- $x_1 > 2 \Rightarrow$ $[(6\ 3)^T, 1]$ $x_1 > 2$
  $[(2\ 7)^T, 0]$ $x_1 = 2$
  $[(9\ 6)^T, 1]$ $x_1 > 2$
  $[(4\ 2)^T, 0]$ $x_1 > 2$

  $\Rightarrow$ avem 3 care respectă
  2 din clasa 1 $(\frac{2}{3})$
  1 din clasa 0 $(\frac{1}{3})$

True

$$G_1 = \sum_{h=1}^{2} \hat{p}_{hk}(1 - \hat{p}_{hk}) = \frac{1}{3}\left(1 - \frac{1}{3}\right) + \frac{2}{3}\left(1 - \frac{2}{3}\right) = \frac{4}{9}$$

clasa 0 clasa 1

False

$$G_2 = \sum_{h=1}^{2} \hat{p}_{hk}(1 - \hat{p}_{hk}) = \frac{1}{1}\left(1 - \frac{1}{1}\right) + \frac{0}{1}\left(1 - \frac{0}{1}\right) = 0$$

clasa 0 clasa 1

- $x_1 > 4$ True : 2 pt clasa 1 $(\frac{2}{2})$ 0 pt clasa 0 $(\frac{0}{2})$ total 2
  False : 0 pt clasa 1 $(\frac{0}{2})$ 2 pt clasa 0 $(\frac{2}{2})$ total 2

$$G_1 = \frac{0}{2}\left(1 - \frac{0}{2}\right) + \frac{2}{2}\left(1 - \frac{2}{2}\right) = 0$$

$$G_2 = \frac{2}{2}\left(1 - \frac{2}{2}\right) + \frac{0}{2}\left(1 - \frac{0}{2}\right) = 0$$

$x_2 > 3$ True: total 2 "0": 1 $(\frac{1}{2})$ "1": 1 $(\frac{1}{2})$

False: total 2 "0": 1 $(\frac{1}{2})$ "1": 1 $(\frac{1}{2})$

$$G_1 = \frac{1}{2} \cdot \left(1 - \frac{1}{2}\right) + \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{2}$$

$$G_2 = \frac{1}{2} \cdot \left(1 - \frac{1}{2}\right) + \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{2}$$

$x_2 > 6$ True: total 1 "0": 1 $(\frac{1}{1})$ "1": 0 $(\frac{0}{2})$

False: total 3 "0": 1 $(\frac{1}{3})$ "1": 2 $(\frac{2}{3})$

$$G_1 = \frac{1}{1}\left(1 - \frac{1}{1}\right) + \frac{0}{1}\left(1 - \frac{0}{1}\right) = 0$$

$$G_2 = \frac{1}{3}\left(1 - \frac{1}{3}\right) + \frac{2}{3}\left(1 - \frac{2}{3}\right) = \frac{4}{9}$$

$\Rightarrow$ cel mai mic Gini index il are $x_1 > 4$ (ambii copii au Gini = 0)

label −1. Which points are the support vectors for a hard-margin

3. Consider training a classification decision tree given a design matrix $\boldsymbol{X} = \begin{bmatrix} 1 & 3 \\ 3 & 5 \\ 2 & 4 \\ 4 & 7 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of $\boldsymbol{X}$, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest entropy: $x_1 > 1$, $x_1 > 2$, $x_2 > 5$, $x_2 > 6$? Motivate your answer.

2.

$[(1 \;\; 3), 1]$ $\quad$ $[(3 \;\; 5), 0]$

$[(2 \;\; 4), 1]$ $\quad$ $[(4 \;\; 7), 0]$

$H_k = H(\hat{p}_k) = -\sum_{k=1}^{2} \hat{p}_{kk} \log \hat{p}_{kk}$

- $x_1 > 1$ ⇒ True 3 "0": 2 → $\frac{2}{3}$ "1": 1 → $\frac{1}{3}$

  False 1 "0": 0 → $\frac{0}{1}$ "1": 1 → $\frac{1}{1}$

$H_1 = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} = 0,27$

$H_2 = \underbrace{-\frac{0}{1}\log 0}_{\text{convention } 0} - \frac{1}{1}\log\frac{1}{1} = 0$

- $x_1 > 2$ ⇒ True : 2 "0": 2 → $\frac{2}{2}$ "1": 0 $\frac{0}{2}$

  False : 2 "0": 0 → $\frac{0}{2}$ "1": 2 $\frac{2}{2}$

$H_1 = -\frac{2}{2}\log\frac{2}{2} - \frac{0}{2}\log\frac{0}{2} = 0$

$H_2 = -\frac{0}{2}\log\frac{0}{2} - \frac{2}{2}\log\frac{2}{2} = 0$

- $x_2 > 5$ True : 1 "0": 1 $\frac{1}{1}$ "1": 0 $\frac{0}{1}$

  False : 3 "0": 1 $\frac{1}{3}$ "1" 2 $\frac{2}{3}$

$H_1 = -\frac{1}{1}\log\frac{1}{1} - \frac{0}{1}\log\frac{0}{1} = 0$

$H_2 = -\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3} = 0,27$

$0 X_2 > 6$ True: 1 "0": 1 $\frac{1}{1}$ "1": 0 $\frac{0}{1}$

False: 3 "0": 1 $\frac{1}{3}$ "1" 2 $\frac{2}{3}$

$$H_1 = -\frac{1}{1}\log\frac{1}{1} - \frac{0}{1}\log\frac{0}{1} = 0$$

$$H_2 = -\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3} = 0,27$$

⇒ smallest entropy : $X_2 > 2$

③.

3. Consider training a classification decision tree given a design matrix $\boldsymbol{X} = \begin{bmatrix} 7 & 1 \\ 5 & 7 \\ 1 & 6 \\ 3 & 9 \\ 2 & 4 \\ 4 & 8 \end{bmatrix}$ and labels $\in \mathbb{R}^{6\times 2}$

$y = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of $\boldsymbol{X}$, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest Gini index: $x_1 > 1$, $x_1 > 3$, $x_2 > 1$, $x_2 > 6$? Motivate your answer.

- $x_1 > 1$

$$G_1 = \frac{2}{5}\cdot\left(1-\frac{2}{5}\right) + \frac{3}{5}\left(1-\frac{2}{5}\right) = \frac{6}{25} + \frac{6}{25} = \frac{12}{25}$$

$$G_2 = \frac{1}{1}\left(1-\frac{1}{1}\right) + \frac{0}{1}\left(1-\frac{1}{1}\right) = 0$$

- $x_1 > 3$

$$G_1 = \frac{1}{3}\left(1-\frac{1}{3}\right) + \frac{2}{3}\left(1-\frac{2}{3}\right) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$$

$$G_2 = \frac{2}{3}\left(1-\frac{1}{3}\right) + \frac{1}{3}\left(1-\frac{1}{3}\right) = \frac{4}{9}$$

- $x_2 > 1$

$$G_1 = \frac{2}{5}\left(1-\frac{2}{5}\right) + \frac{3}{5}\left(1-\frac{3}{5}\right) = \frac{12}{25}$$

$$G_2 = \frac{1}{1}(1-1) + \frac{0}{1}\left(1-\frac{0}{1}\right) = 0$$

- $x_2 > 6$

$$G_1 = \frac{0}{3}\left(1-\frac{0}{3}\right) + \frac{3}{3}\left(1-\frac{3}{3}\right) = 0$$

$$G_2 = \frac{3}{3}\left(1-\frac{3}{3}\right) + \frac{0}{3}\left(1-\frac{0}{3}\right) = 0$$

$\Rightarrow$ smallest Gini index $x_2 > 6$

4. Consider a training classification **decision tree** given a design matrix X = [6 3; 2 7; 9 6; 4 2] and labels y = [1; 0; 1; 0]. Let x1 denote feature 1, corresponding to the first column of X and let x2 denote feature 2, corresponding to the second column of X, Which of the following splits at the root node gives the smallest entropy: x1 > 2, x1 > 4, x2 > 3, x2 > 6? Motivate your answer

Consider training a classification decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of X, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest entropy: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

- $x_1 > 2$: $H_1 = -\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3} = 0{,}27$

  $H_2 = -\frac{1}{1}\log\frac{1}{1} - \frac{0}{1}\log\frac{0}{1} = 0$

- $x_1 > 4$: $H_1 = -\frac{0}{2}\log\frac{0}{2} - \frac{2}{2}\log\frac{2}{2} = 0$

  $H_2 = -\frac{2}{2}\log\frac{2}{2} - \frac{0}{2}\log\frac{0}{2} = 0$

- $x_2 > 3$: $H_1 = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2} = 0{,}3$

  $H_2 = -11- = 0{,}3$

- $x_2 > 6$: $H_1 = -\frac{1}{1}\log\frac{1}{1} - \frac{0}{1}\log\frac{0}{1} = 0$

  $H_2 = -\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3} = 0{,}27$

$\Rightarrow$ smallest entropy $x_1 > 4$